

transit-lab: Technical Report

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A Derivations

A.1 Conventions

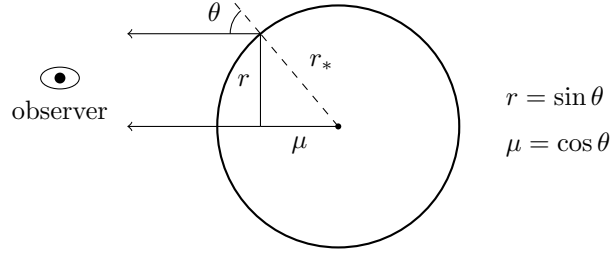


Figure 1: Side view of the star, sliced through its centre, with the observer to the left. A point on the stellar surface is at angle θ between its surface normal and the line of sight; its projected distance from the centre is r , and μ is the corresponding depth coordinate.

With the stellar radius normalised to 1,

$$r = \sin \theta, \quad \mu = \cos \theta. \quad (1)$$

- μ is for the physics of limb darkening: it determines at what depth the photons we receive are coming out.
- r is needed for the geometry: it is how far from the stellar centre the light is coming from.

Let d be the distance between the centres of the star and the planet, r_* the stellar radius and r_p the planetary radius. Define

$$z = \frac{d}{r_*}, \quad p = \frac{r_p}{r_*}. \quad (2)$$

Flux is energy per unit time per unit detector area. The most flux the observer receives is out of transit; when a transit occurs, the flux drops. For light curves we record the normalised flux F , the received flux divided by the out-of-transit flux.

A.2 Occulted area of two overlapping discs

Uniform source. This assumes the whole area of the star emits light equally, so

$$F = 1 - \frac{\text{area of overlap}}{\pi r_*^2}. \quad (3)$$

Everything in the model is normalised, so only the relative size of the two bodies (p) and their relative separation (z) matter. We can therefore write

$$F^e(p, z) = 1 - \lambda^e(p, z), \quad (4)$$

where e stands for eclipse and λ^e is the overlap function.

Case 1: no overlap. There is no overlap if

$$d > r_p + r_* \iff zr_* > pr_* + r_* \iff 1 + p < z. \quad (5)$$

So $\lambda^e(p, z) = 0$ and $F^e(p, z) = 1$ here.

Case 2: planet fully inside the star. This happens when

$$d \leq r_* - r_p \iff zr_* \leq r_* - pr_* \iff z \leq 1 - p, \quad \text{with } p < 1. \quad (6)$$

Then

$$\frac{\text{area of overlap}}{\pi r_*^2} = \frac{\pi r_p^2}{\pi r_*^2} = p^2, \quad (7)$$

so $\lambda^e(p, z) = p^2$ and $F^e(p, z) = 1 - p^2$.

Case 3: star fully inside the planet. This happens when

$$d \leq r_p - r_* \iff z \leq p - 1, \quad \text{with } p \geq 1. \quad (8)$$

Then

$$\frac{\text{area of overlap}}{\pi r_*^2} = \frac{\pi r_*^2}{\pi r_*^2} = 1, \quad (9)$$

so $\lambda^e(p, z) = 1$ and $F^e(p, z) = 0$.

Case 4: partial overlap. This is the most thorough case, and whether $p > 1$ or $p \leq 1$ does not matter. It is the only remaining case, so

$$\underbrace{z > 1 - p > 0}_{p < 1} \quad \text{and} \quad \underbrace{z > p - 1 \geq 0}_{p \geq 1} \quad \text{and} \quad z \leq 1 + p. \quad (10)$$

The first two conditions combine into $z > |1 - p|$, hence

$$|1 - p| < z \leq 1 + p. \quad (11)$$

There are three configurations, all sharing the decomposition $S_1 + S_2 + S_3$ (Figures 2–4).

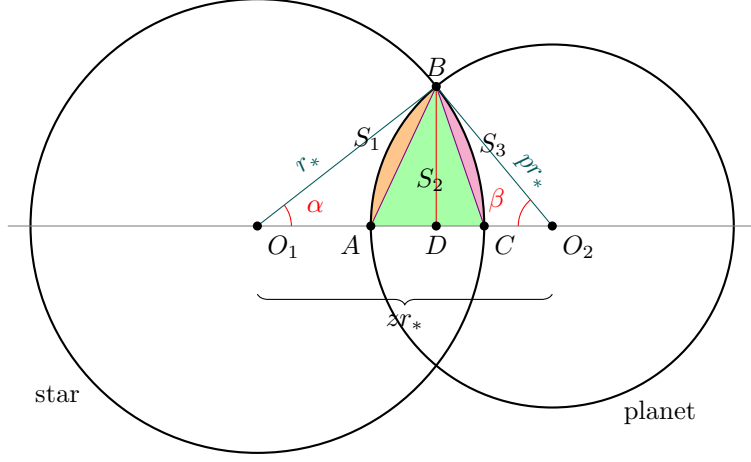


Figure 2: Case 4, configuration 1: α and β both acute (drawn with $p = 0.8$, $z = 1.3$). The star (centre O_1 , radius r_*) and the planet (centre O_2 , radius pr_*) have centres a distance zr_* apart. B is an intersection point of the two circles; A is where the planet's boundary meets the axis O_1O_2 inside the star; C is where the star's boundary meets the axis inside the planet; D is the foot of the perpendicular from B . The upper half of the overlap is split into the planet segment S_1 on chord AB , the triangle $S_2 = ABC$, and the star segment S_3 on chord BC .

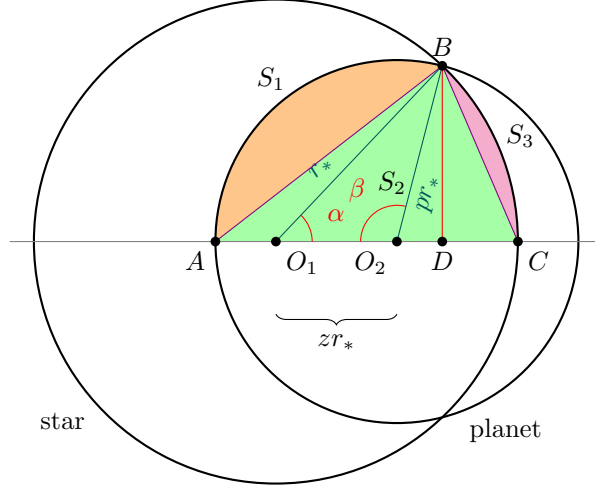


Figure 3: Case 4, configuration 2: β obtuse, with the planet's centre O_2 inside the star (drawn with $p = 0.75$, $z = 0.5$).

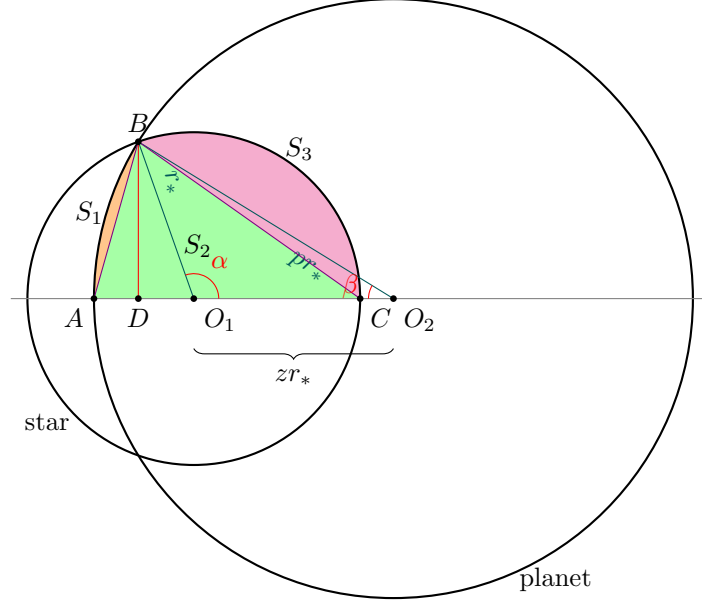


Figure 4: Case 4, configuration 3: $p > 1$ and α obtuse (drawn with $p = 1.8$, $z = 1.2$).

Let $\alpha = \angle BO_1O_2$ and $\beta = \angle BO_2O_1$. By the cosine rule in triangle O_1BO_2 :

$$(pr_*)^2 = r_*^2 + z^2r_*^2 - 2zr_*^2 \cos \alpha \quad \implies \quad \cos \alpha = \frac{z^2 - p^2 + 1}{2z}, \quad (12)$$

$$r_*^2 = p^2r_*^2 + z^2r_*^2 - 2p zr_*^2 \cos \beta \quad \implies \quad \cos \beta = \frac{z^2 + p^2 - 1}{2pz}, \quad (13)$$

so

$$\alpha = \cos^{-1}\left(\frac{z^2 - p^2 + 1}{2z}\right), \quad \beta = \cos^{-1}\left(\frac{z^2 + p^2 - 1}{2pz}\right). \quad (14)$$

The height of B above the axis is $BD = r_* \sin \alpha$, and

$$AC = zr_* - (zr_* - pr_*) - (zr_* - r_*) = r_*(1 + p - z). \quad (15)$$

The three areas are

$$S_2 = \frac{1}{2} BD \cdot AC = \frac{1}{2} r_*^2 \sin \alpha (1 + p - z), \quad (16)$$

$$S_1 = \pi p^2 r_*^2 \frac{\beta}{2\pi} - \frac{1}{2} p^2 r_*^2 \sin \beta = \frac{1}{2} p^2 r_*^2 (\beta - \sin \beta), \quad (17)$$

$$S_3 = \frac{1}{2} r_*^2 (\alpha - \sin \alpha). \quad (18)$$

By symmetry about the axis, the overlap area is $2(S_1 + S_2 + S_3)$. Hence

$$\lambda^e(p, z) = \frac{2(S_1 + S_2 + S_3)}{\pi r_*^2} = \frac{2}{\pi} \left[\frac{1}{2} p^2 (\beta - \sin \beta) + \frac{1}{2} (\alpha - \sin \alpha) + \frac{1}{2} \sin \alpha (p - z + 1) \right]. \quad (19)$$

Using

$$\sin \alpha = \sqrt{1 - \cos^2 \alpha}, \quad \sin \beta = \frac{\sin \alpha}{p} \quad (20)$$

(the latter from the sine rule in triangle O_1BO_2), this becomes

$$\lambda^e(p, z) = \frac{2}{\pi} \left[\frac{1}{2}p^2\beta + \frac{1}{2}\alpha - \frac{1}{2}p \sin \alpha + \frac{1}{2} \sin \alpha (p - z) \right] \quad (21)$$

$$= \frac{2}{\pi} \left[\frac{1}{2}p^2\beta + \frac{1}{2}\alpha - \frac{1}{2}z \sin \alpha \right] \quad (22)$$

$$= \frac{2}{\pi} \left[\frac{1}{2}p^2\beta + \frac{1}{2}\alpha - \frac{1}{2} \sqrt{\frac{4z^2 - (z^2 - p^2 + 1)^2}{4}} \right] \quad (23)$$

$$= \frac{1}{\pi} \left[p^2\beta + \alpha - \sqrt{\frac{4z^2 - (z^2 - p^2 + 1)^2}{4}} \right]. \quad (24)$$

Result. Hence we get

$$\lambda^e(p, z) = \begin{cases} 0, & 1 + p < z, \\ \frac{1}{\pi} \left[p^2\beta + \alpha - \sqrt{\frac{4z^2 - (1 + z^2 - p^2)^2}{4}} \right], & |1 - p| < z \leq 1 + p, \\ p^2, & z \leq 1 - p, \\ 1, & z \leq p - 1, \end{cases} \quad (25)$$

with α and β as above.

A.3 Quadratic limb darkening

This model is much more realistic. Stars appear darker towards their edge (this is limb darkening), because near the limb our line of sight only reaches cooler outer layers of the star. Here, I describe the brightness across the disc with the standard quadratic law,

$$I(r) = 1 - \gamma_1(1 - \mu) - \gamma_2(1 - \mu)^2, \quad \mu = \sqrt{1 - r^2}. \quad (26)$$

This is the simplest curve that equals 1 at the centre and has enough freedom to fit real stars. The coefficients γ_1 and γ_2 will be taken from published tables.

Setting. Figure 5 shows the setting.

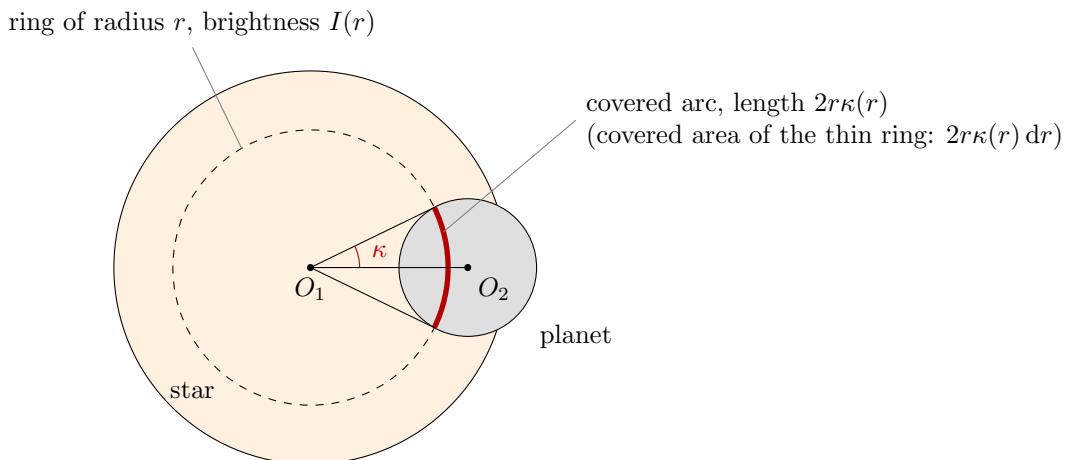


Figure 5: A ring of radius r on the stellar disc, of uniform brightness $I(r)$. The planet covers an arc of the ring subtending the half-angle $\kappa(r)$ at the stellar centre O_1 .

Brightness. Brightness, or specific intensity, is

$$I_\nu = \frac{dE}{dt \cdot dA \cos \theta \cdot d\Omega \cdot d\nu}, \quad (27)$$

where

- dE is the energy;
- dt is the time interval;
- $dA \cos \theta$ is the projected area perpendicular to the ray, where θ is the angle between the surface normal and the ray;
- $d\Omega$ is the solid angle (the direction of the ray);
- $d\nu$ is the frequency interval.

Assumption. For limb darkening, the brightness at the same radius r from the centre is equal. So we can use the concept of a ring to compute the overall brightness when an eclipse occurs.

Derivation. We have

$$F(p, z) = 1 - \frac{\text{blocked brightness}}{\text{total brightness}}. \quad (28)$$

We will use an integral where the radius r slides from 0 to 1, and we calculate the brightness blocked by the overlap on the ring at radius r from the centre and of width dr . Since the total brightness comes from full rings, each of length $2\pi r$,

$$\text{total brightness} = \int_0^1 I(r) 2\pi r dr. \quad (29)$$

Now, let $\kappa(r)$ be the function returning the half-angle of the ring fraction that holds the overlap at radius r (Figure 5). Then

$$\text{blocked brightness} = \int_0^1 I(r) 2\pi r \cdot \frac{2\kappa(r)}{2\pi} dr = \int_0^1 I(r) 2r\kappa(r) dr. \quad (30)$$

Geometrically, from the cosine rule in the triangle formed by O_1 , O_2 and an endpoint of the covered arc,

$$\kappa(r) = \begin{cases} \pi, & r \leq p - z \quad (\text{ring entirely inside the planet}), \\ \cos^{-1}\left(\frac{r^2 + z^2 - p^2}{2rz}\right), & |z - p| < r < z + p \quad (\text{ring crosses the planet's edge}), \\ 0, & \text{otherwise} \quad (\text{ring misses the planet}). \end{cases} \quad (31)$$

Hence we get our final relative flux formula:

$$F(p, z) = 1 - \frac{\int_0^1 I(r) 2r\kappa(r) dr}{\int_0^1 I(r) 2\pi r dr}. \quad (32)$$

A.4 Orbital separation

Suppose $\omega = 2\pi/T$ is the orbital angular frequency (T is the period), a is the orbital radius (circular orbit), and t is the time measured from mid-transit. What is $z(t)$? We need it to plot the light curve, the relative flux $F(p, z)$ against time.

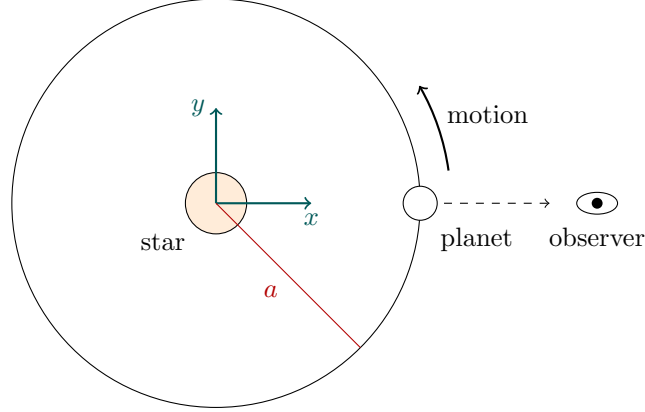


Figure 6: Top view of the orbit at mid-transit ($t = 0$). The observer lies on the x -axis, and the planet moves up (towards $+y$).

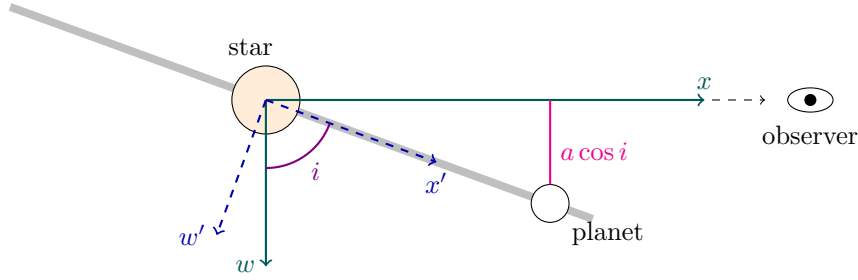


Figure 7: Edge-on view of the tilted orbit at mid-transit (drawn with $i = 70^\circ$). The orbit lies in the x' - y plane, rotated about the y -axis (pointing out of the page); i is the angle between the orbit normal w' and the line of sight x , equivalently between x' and w .

I will operate with the real distance $d = zr_*$ for the derivation, then divide by r_* to get z .

Edge-on orbit. First, let us consider the simplest case, $i = 90^\circ$. Assume the planet in Figure 6 moves up, so if we plot its coordinates in the x - y plane, we get

$$x(t) = a \cos\left(\frac{2\pi}{T}t\right) = a \cos(\omega t), \quad (33)$$

$$y(t) = a \sin\left(\frac{2\pi}{T}t\right) = a \sin(\omega t), \quad (34)$$

and, since the observer looks along the x -axis,

$$d = |y(t)| = a |\sin(\omega t)|. \quad (35)$$

Inclined orbit. Now we add the tilt (a rotation about the y -axis), which introduces the third coordinate, w (Figure 7). In the (x', y, w') system, the coordinates are $(a \cos(\omega t), a \sin(\omega t), 0)$. We need to convert them to (x, y, w) . Since the observer sits on the x -axis, that coordinate is irrelevant, so the answer will be $d = \sqrt{y^2 + w^2}$. The conversion gives

$$x' \rightarrow x : a \cos(\omega t) \rightarrow a \cos(\omega t) \sin i, \quad (36)$$

$$w' \rightarrow w : 0 \rightarrow a \cos(\omega t) \cos i. \quad (37)$$

Therefore

$$d(t) = \sqrt{a^2 \sin^2(\omega t) + a^2 \cos^2(\omega t) \cos^2 i} \quad (38)$$

and

$$z(t) = \frac{a}{r_*} \sqrt{\sin^2(\omega t) + (\cos(\omega t) \cos i)^2}. \quad (39)$$

Note. This function gives exact values (including those for a transit) twice per period. As we do not want to record a dip in flux when the planet is *behind* the star, we apply the constraint $\cos(\omega t) > 0$.

A.5 Transit duration

A.6 The box-least-squares statistic

A.7 Period-grid spacing

A.8 Transit signal-to-noise